NOT MEASUREMENT SENSITIVE

MIL-T-62097C
10 October 1991
SUPERSEDING
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29 September 1982

MILITARY SPECIFICATION

TRUCK, MOTOR HOME SHELL, FOR CUSTOMIZATION INTO MOBILE LABORATORY, ELECTRONICS VAN, WORK AREA, ETC, 14,000 POUNDS GVW, 4x2, MODIFIED COMMERCIAL

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers diesel and gasoline engine driven four-wheel, two-rear wheel drive, forward control motor home shells, having a minimum gross vehicle weight (GVW) of 14,000 pounds, for normal operating conditions. The vehicle has an integral driver's compartment and body for later installation of such special equipment as may be desired. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in procurement documents.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-2320

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SPECIFICATIONS

FEDERAL

	W-B-131	- Battery, Storage: Vehicular, Ignition, Lighting and Starting.
*	A-A-393	- Extinguishers, Fire, Dry-Chemical (Hand Portable).
*	VV-F-800	- Fuel Oil, Diesel.

MILITARY

	MIL-P-514	- Plate, Identification, Instruction And Marking, Blank.
*	MIL-T-5624	- Turbine Fuel Aviation, Grades JP-4 and JP-5.
	MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
*	MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.

STANDARDS

FEDERAL

*	FED-STD-297	- Rustproofing of Commercial (Nontactical) Vehicles.

MILITARY

	MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
	MIL-STD-1223	- Non-Tactical Wheeled Vehicles Treatment, Painting, Rustproofing, Identification Marking, and Data Plate Standards.
*	MIL-STD-1791	- Designing for Internal Aerial Delivery in Fixed Wing Aircraft.

Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specification and Standards, Bldg. 4D, 700 Robbins Avenue Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

* DEPARTMENT OF DEFENSE (DOD)

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

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* AFSC DESIGN HANDBOOK
DH-1-11 - Air Transportability Handbook

(Application for copies of the AFSC Design Handbook should be addressed to Wright-Patterson AFB, ASD/ENESS, Mr. Rod Wright, Wright-Patterson AFB, OH 45433, telephone (513) 255-6281.)

DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should reference the Code of Federal Regulations, 49 CFR, and the Federal Register, and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
ASTM D 4985 - Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

(Application for copies of ASTM publications should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

	NFPA No. 70	- National Electrical Code.

* (Application for copies of NFPA publications should be addressed to the National Fire Protection Association, 60 Batterymarch Park, Quincy, MA 02269.)

SAE, INC.

SAE Standards and Recommended Practices

*	J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD Adopted).
	J537	- Storage Batteries.
*	J551	- Limits and Methods of Measurement of Radio Interference Characteristics of Vehicles and Devices (30-1000 MHz) (DOD adopted).
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection.
	J688	- Truck Ability Prediction Procedure.
*	J1349	- Engine Power Teat Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC.

Year Book.

* (Application for copies of the Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.13, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification.

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Except as specified in 3.1.1 through 3.1.1.13, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis model components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data and the body and mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering offices of the procuring activity, prior to the delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

* 3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

* 3.1.1.2 Marking and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification marking and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2) concealed markings shall be furnished.

* 3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Drain plugs. All drain plugs installed in manual transmissions and rear axles shall be of the permanent magnet type.

3.1.1.5 Wood treatment. When specified (see 6.2), wood shall be treated in accordance with MIL-STD-1223.

* 3.1.1.6 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the chassis manufacturer's standard single center mounted eye or pin shall be mounted on the front of the vehicle.

* 3.1.1.7 Spark arrester. As specified herein (see 3.4.4), the vehicle shall be furnished with an exhaust system spark arrester.

* 3.1.1.8 Air transportability. When specified (see 6.2), the vehicle shall be air transportable in C-5A aircraft in accordance with the requirements of MIL-STD-1791 and AFSC Design Handbook DH-1-11. Removal or relocation of mechanically attached (nonwelded, nonriveted, etc.) components with hand tools, requiring not more than 1 manhour total to

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remove, relocate and tiedown; and not more than 1 manhour total to return the vehicle to it's original, as opposed to reduced, configuration; shall be acceptable. The self-mobility of the vehicle shall not be affected by reducing its configuration. Tiedowns for removed or relocated equipment shall be furnished. The curb weight of the vehicle shall not exceed 5,000 pounds on the front axle and 10,000 pounds on the rear axle. In addition to the requirements of 3.2.6, the rated capacity of the axles and suspension system shall not be less than 1-1/4 times the load imposed on each by the curb weight of the vehicle. The vehicle shall be air transportable as described above without any other special provisions and without any shoring. The vehicle shall not be delivered to the Government in its reduced configuration. Drawings and data will be required to be submitted by the contractor. Government approval of the data (120 days) and then comparison of the vehicle to the approved data will be required before acceptance of the vehicle.

* 3.1.1.9 Lifting and tiedown attachments. When specified (see 6.2), the vehicle shall be equipped with lifting and tying down attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. In MIL-STD-209G, replace "maximum shipping weight (MSW)" wherever it appears with "curb weight." A transportation plate conforming to composition A (class 1 or 2) or composition C of MIL-P-514 shall be provided. The transportation plate shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and length of each sling cable. A silhouette of the vehicle showing the center of gravity shall be provided on the transportation plate. Tying down attachments shall be identified by stenciling or other suitable marking. Tie down markings shall clearly indicate that the attachments are intended for the tiedown of the equipment on the carrier,

3.1.1.10 Blackout curtains. When specified (see 6.2), blackout curtains or screens shall be provided to allow operation, during blackout conditions, of all the interior dome lights and fluorescent lights to the rear of the driver's seat. Full height transverse blackout screens or curtains shall be provided to cover the full width of the interior at the rear of the driver's seat. Blackout curtains or screens shall be provided for all exterior windows to the rear of the driver's seat.

* 3.1.1.11 Wheel splash and stone throw protection. Rear end mud flaps shall be provided. A metal strip not less than 0.125 inch thick and not less than 1 inch wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 1 inch high by 1.25 inch wide by 0.094 inch thick shall be furnished at each bolt. Mud flap installations shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

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3.1.1.12 Silicone brake fluid. When specified (see 6.2), and if available as the manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A decal or plate shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

* 3.1.1.13 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.2.2 Air pollution control. The vehicle shall comply with Environmental Protection Agency Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3..2.3 Sound level. The interior sound level shall not exceed 84 db(A), when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to EPA Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, equipment, body; and a full complement of fuel, lubricants, and coolant.

3.2.5 Gross vehicle weight. Gross vehicle weight (GVW) shall consist of curb weight, operator (weight computed at 175 pounds), and a payload to provide a total GVW of not less than 14,000 pounds.

3.2.6 Weight distribution. The distribution of the GVW for the purpose of establishing suspension, axle and tire capacity shall be determined with the payload uniformly distributed over the payload area. Rated capacities shall be measured at the ground.

3.2.7 Wheelbase. The wheelbase shall be not less than 178 inches, measured between the center lines of the front and rear axles. When specified (see 6.2), wheelbase shall be compatible with the body make and model specified.

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3.2.8 Ratings. Vehicle ratings shall be manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings shall be submitted to the engineering office of the procuring activity.

* 3.2.9 Overall width. The overall width of the vehicle, exclusive of tires and safety related items such as mirrors, lights and reflectors, shall be not more than 96 inches. The width over the tires shall be not more than 100 inches.

* 3.2.10 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance to other compartments and systems.

* 3.2.11 Prohibited materials. Asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance.

3.3.1 Speeds. High and low speed requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend a 2.2 percent grade at 50 miles per hour (mph). Gradeability requirements shall be met with the main transmission in direct drive. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

3.3.1.2 Low speed. Low speed shall be calculated with the engine operating at not less than 35 percent of the recommended governed speed, and shall provide a vehicle speed of not more than 4.5 mph.

* 3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 58 mph. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{Governed speed (rpm)}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

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3.4 Chassis components.

3.4.1 Engine. The engine furnished shall be the chassis manufacturer's standard or optional engine for the commercial model vehicle which meets or exceeds the requirements of this specification.

* 3.4.1.1 Diesel engine. Unless otherwise specified, (see 3.4.1.2) the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower ratings used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine with not less than six cylinders. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed not more than the manufacturer's recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel, when used in accordance with the operator's manual. Net horsepower ratings used in performance calculations shall be determined in accordance with SAE J1349. Unless otherwise required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline.

* 3.4.1.3 Oil filter. A full flow or combination full flow and bypass oil filter with replacement element shall be furnished.

3.4.1.4 Governor. An engine governor shall be furnished and set and sealed to limit the engine to the engine manufacturer's maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain the engine coolant at a temperature below the boiling point with the vehicle loaded to GVW and operated at an altitude of 10,000 feet above sea level or in an ambient air temperature of not less than 125 degrees Fahrenheit (ºF). The cooling system shall include a surge tank or a coolant recovery reservoir of not less than two-quart capacity. For cooling system servicing, see 3.4.28.1.

* 3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

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3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Power plant heaters and fuel warmer. When specified (see 6.2), a coolant heater, an engine oil heater, and fuel warmers (diesel engine driven vehicles only) shall be provided. The heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof, slave receptacle for receiving external power and for grounding the vehicle. A three-wire connecting cable, not leas than 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). The electrical insulation of the connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -60ºF) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable assembly shall be provided. Heaters and fuel warmers shall be furnished as follows:

* (a) A coolant heater, 1000-watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. An engine thermostat with an operating range of 170ºF to 195ºF shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements shall be installed:
  (1) Not less than 15 watts per quart heating capacity
  (2) Not more than 18 watts per square inch heating capacity
  (3) Thermal balance design or thermostat control providing for uninterrupted operation
  (4) Provision for mounting below minimum service oil level.

* (c) An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40ºF to an outlet temperature of +9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 80ºF under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 80°F under any possible

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condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.8(c) and (d) may be combined.

* 3.4.1.9 Fuel fired engine preheater. When specified (see 6.2), for diesel engine driven vehicles, a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and a circulating pump and shall be connected to the engine coolant system. The heater shall be capable of starting and operating at -60ºF and shall heat the engine to +40°F from -60°F in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating. The system shall include fuel warmers or preheaters conforming to 3.4.1.8(c) and (d).

* 3.4.1.10 Vernier throttle control. When specified (see 6.2), a manual vernier controlled throttle with quick release shall be furnished, except on electronically controlled engines.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or electric grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished. When specified (see 6.2) for diesel engines, the starting motor shall be equipped with a thermostat controlled, automatic resetting circuit breaker to protect the motor from overcrank heat damage. The circuit breaker shall not limit cranking ability for a startable engine.

* 3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.

* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2), on diesel engine driven vehicles a minimum 65-ampere alternator shall be furnished. For diesel engine vehicles, the alternator output at engine idle speed shall be not less than 45 amperes. Unless otherwise specified (see 6.2), on gasoline engine driven vehicles, an alternator of not less than 85 amperes rated capacity and which provides not less than 45 amperes dc output at normal engine idle speed, shall be furnished.

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3.4.2.4 Lighting. All vehicle lights, reflectors, and wiring shall be as specified herein and shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails (unless recessed and fully protected) or mounted on vehicle bumpers. Taillights, stoplights and rear turn signals shall be located to avoid obstruction by the rear doors when the doors are in the full open position.

* 3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. The vehicle shall have lights integrally mounted in the body. Turn signal operating units shall have a visible flash indicator.

* 3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ratings at 0ºF, both measured in accordance with SAE J537, shall be not less than specified in table I. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

TABLE I. Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Diesel	480	1875
Gasoline	100	450

* 3.4.2.7 Radio interference suppression. The vehicle shall be suppressed to limit electromagnetic radiation in accordance with SAE J551. Any body equipment emitting radiation shall be suppressed to the same level as the vehicle chassis.

3.4.2.8 Fog lamps. When specified (see 6.2), two sealed beam fog lamps shall be furnished and installed on the front of the vehicle in a protected location. A fog lamp switch shall be located on the vehicle instrument panel.

3.4.3 Fuel system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

* 3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank. Unless otherwise specified (see 6.2), the fuel tank shall be manufacturer's standard with not less than 50 gallons total capacity (43 gallons for vehicles destined for California). When more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure equalized fuel level in both tanks. When more than one tank is

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furnished on gasoline engine driven vehicles, a selector valve connecting either tank to engine fuel intake shall be provided and means shall be provided to monitor the level of either tank from a single fuel gage; or an equalizing pump shall be used to maintain the same fuel level in both tanks.

* 3.4.3.3 Fuel and water separator. The manufacturer's standard or optional fuel filter shall be provided. When specified (see 6.2), a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve. When a separator is required, a combination filer/separator unit may be furnished.

* 3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. The exhaust system shall discharge at the rear of the vehicle or to the side behind the rear wheels. Except when a turbocharged engine is specified, a spark arrester shall be furnished. The spark arrester shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.4.5 Tranamission. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. Transmission and axle gear ratios shall be selected to provide the performance specified in 3.3.1 through 3.3.1.3.

* 3.4.5.1 Automatic transmission. The vehicle shall have an automatic transmission. The automatic transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with a power takeoff opening.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.4.7 Frame. The chassis frame shall be the manufacturer's standard for the model vehicle furnished. Frame members shall provide sufficient structural strength in the chassis frame through increased resisting bending moment to at least equal the load imposed with the vehicle loaded to specified GVW. Chassis frame rails shall not protrude beyond the rear end of the vehicle.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the vehicle loaded to specified GVW. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Vehicle shall be equipped with heavy duty hydraulic, double-acting shock absorbers at the front and rear wheels. The suspension system shall resist vibration, road noise, tip and roll of the body. When specified (see 6.2), variable rate springs and adjustable shock absorbers shall be provided.

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* 3.4.9 Axles. Axle ratings shall be at least equal to the load imposed on each axle measured at the ground, with the vehicle loaded to specified GVW. A single speed, single reduction rear axle shall be furnished. Wheel bearings and axle spindles shall be oil lubricated. The hubcaps, except for driving axles, shall have a window for visual determination of oil level. Provisions for venting or withstanding internal pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Traction control. When specified (see 6.2), the vehicle shall be furnished with a traction control on the rear axle. The traction control shall actuate automatically to ensure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires, and tubes. The vehicle shall be equipped with single front and Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in procurement documents, multi-piece rims shall not be furnished. Rim and disc wheel sizes shall be the same for all wheels on each vehicle. Tire size and load range (ply rating) shall be the same for all tires on each vehicle. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Unless otherwise specified (see 6.2), tires shall be tubeless type with highway tread. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be of the steel belted radial type or when specified (see 6.2), bias ply. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel, at the ground, with the vehicle loaded to the specified GVW. Tires and tire size designation systems shall conform to the Tire and Rim Association or to the European Tyre and Rim Technical Organization recommendations.

3.4.10.2 Inner tubes. When tube type tires are specified in procurement documents, inner tubes shall be of the heavy duty type, and shall be of the proper size for the tires furnished. Tire flaps shall be provided for tube type tires in accordance with the Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. When specified (see 6.2), a carrier for a spare wheel or rim and tire assembly shall be installed in a readily accessible location on the vehicle. Threaded fasteners, when used to secure the spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. The carrier design shall enable safe removal or mounting of the spare wheel assembly using only tools specified in 3.4.15.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and

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carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

* 3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare wheel or rim or spare tire assembly is furnished on a vehicle without a carrier, it shall be stowed securely on the vehicle for shipment. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

* 3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.42 and 393.45 through 393.52. Brake linings shall be nonasbestos material.

3.4.11.1 Power-hydraulic type. The vehicle shall be equipped with power assisted, hydraulically actuated, four-wheel service brakes.

* 3.4.11.1.1 Split hydraulic brake system. The power-hydraulic brake system shall be so arranged as to provide separate systems for at least two wheels and so designed and constructed that rupture or leakage type failure of any single pressure component of the service brake system, except structural failures of the brake master cylinder body, effectiveness indicator body, or other housing common to the divided system, will not result in complete loss of function of the vehicle brakes when force on the brake pedal is continued. Pressure component means any internal component of the brake master cylinder or master control unit, wheel brake cylinder, brake line, brake hose, or equivalent, except power assist components.

* 3.4.11.1.2 Indicator light. The split hydraulic brake system shall be equipped with an electrically operated red light mounted on the instrument panel to indicate system effectiveness. The light shall have an area of not less than 0.196 square inch. It shall illuminate before or upon application of the brakes when an actuating pressure component of the system has sustained a loss of pressure. The indicator light system shall include a means for testing by the vehicle operator to assure that the light bulb is operable.

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3.4.11.2 Parking brake. A manually operated parking brake shall be provided. The parking brake shall be capable of actuating the vehicle rear brakes. The hand control shall be located in an accessible location for the driver. The hand control shall be capable of manual adjustment.

3.4.12 Power steering. Power steering shall be furnished.

* 3.4.13 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type and operated by either air or electric motor(s). When specified (see 6.2), and if electric motor wipers are furnished, intermittent wipers shall be furnished.

3.4.14 Bumpers. Unless the bumper is an integral part of the vehicle, a channel type front bumper shall be provided.

3.4.14.1 Rear end protection. The rear end of the vehicle shall be protected in accordance with Federal Motor Carrier Safety Regulation 393.86. The rear bumper shall be provided as specified herein.

* 3.4.15 Tool stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, antiskid chains (for outside tires on duals only) and emergency reflective triangles shall be furnished. The stowage space shall provide for positive retainment of this equipment during vehicle operation. Stowage space for these tools may be furnished in the cab. When stowage space for these tools is located outside of the cab, it shall be weatherproof and shall provide for locking with a padlock.

3.4.15.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a hydraulic jack, jack handle and wheelnut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel and tire assembly.

3.4.16 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

3.4.17 Ventilator. Ventilator(s), of the static type, capable of providing an adequate supply of fresh air and of properly ejecting foul air under all conditions of operation shall be provided. Vent openings shall be provided with a weatherproof cover with means for opening and closing the cover from inside the vehicle. Outside openings for heaters and ventilators shall be so located as to minimize entry of fumes, rain, road dust, and road wash.

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3.4.18 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard air conditioning system, including tinted windshield and tinted glass. When specified (see 6.2), an air conditioning system of increased capacity shall be provided.

3.4.19 Engine compartment cover. An engine compartment insulated cover, for engine and component maintenance, shall be manufacturer's standard type. When specified (see 6.2), the engine compartment access cover shall be airtight, shall be heavily insulated for soundproofing, or shall be customized as required.

* 3.4.20 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be visible to the driver when seated in the driving position.

* 3.4.21 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Inside rearview mirror
(b) Key-operated ignition switch
(c) Ammeter or voltmeter, or charging indicator
(d) Fuel gage
* (e) Oil pressure gage or red indicator warning light
* (f) Engine coolant temperature gage or high coolant temperature or low coolant level red indicator warning light
* (g) High coolant temperature or low coolant level warning buzzer
(h) Speedometer with recording odometer
(i) Ash receptacle
(j) Dual sunvisors
(k) Front door mounted armrests on driver's and passenger's sides
(l) Driver's compartment ventilator other than window
(m) Map reading lamp, shaded
(n) Tachometer (for diesel engine driven vehicles).

When specified (see 6.2), an automatic engine shutdown system also shall be furnished. The system shall include an engine coolant temperature, engine coolant level and engine oil pressure red indicator warning light and alarm buzzer. The warning light and buzzer actuation shall preceed engine shutdown. The system shall permit engine restart and run for approximately 30 seconds following automatic shutdown. Also, engine oil pressure and engine coolant temperature gages shall be provided with the automatic engine shutdown system.

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3.4.22 Rearview mirrors. Outside rearview mirrors shall be mounted on each side of the vehicle. The mirrors shall have flat and convex areas. The flat portion shall have not less than 50 square inches of reflective area. The convex portion shall have not less than 24 inches of reflective area with a radius of curvature of not less than 20 inches. The convex portion shall be attached to the lower mirror supporting arm and shall not interfere with use of the flat mirror. The mirrors shall have not less than two supporting arms. When specified (see 6.2), the curbside flat mirror shall be of the motorized type, with remote control. The mirror motor shall provide not less than 60 degrees horizontal rotational viewing range. When specified (see 6.2), the flat mirrors shall be electrically heated. Mirror remote and heating controls shall be within reach of the seated driver.

3.4.23 Additional accessories. When specified (see 6.2), additional accessories, such as a vehicle AM-FM or a two-way transmission radio in the driver's compartment or any other special accessories, shall be provided as required.

3.4.24 Horn. The manufacturer's standard electric horn shall be furnished.

3.4.25 Insect screen. When specified (see 6.2), an insect screen shall be provided on the front of the radiator. The screen shall be corrosion-resistant.

3.4.26 Leveling jacks. When specified (see 6.2), hydraulic leveling jacks at the four corners of the chassis frame shall be furnished. The leveling jacks shall be individually operated by controls in the driver's compartment. Each control shall be capable of raising or lowering a jack to level the chassis frame on inclined and level ground surfaces.

3.4.27 Engine hour meter. When specified (see 6.2), an engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.28 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine and brake systems; filling and charging of batteries; alinment of front wheels; inflation of all tires; complete lubrication of chassis, engine and running gear with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of the cooling system in accordance with 3.4.28.1; and servicing of windshield washer reservoir with water and appropriate additives.

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* 3.4.28.1 Engine coolant. The engine coolant shall be a solution of ethylene glycol antifreeze and water or propylene glycol antifreeze and water, in equal parts of antifreeze and water by volume. Ethylene glycol antifreeze shall conform to ASTM D4985 with not more than 250 parts per million silicates. Propylene glycol antifreeze shall be in the specific formulation approved by the engine and cooling system manufacturers. When specified (see 6.2), the percentage of antifreeze in the cooling system shall be increased to provide protection against freezing down to -65°F.

* 3.5 Body. The body shall be built on a forward control chassis and shall have an integral driver's compartment and enclosed body. An upholstered, bucket type driver's seat with pedestal shall be furnished. The seat shall be adjustable fore and aft and for height. The driver's seat pedestal shall be provided with a positive locking mechanism to secure the seat in place when in use. Body mounting shall be in accordance with the latest edition of the chassis manufacturer's body builders layout. The body shall be the manufacturer's standard model size unit utilizing a minimum 178 inch wheelbase (see 3.2.7). When specified (see 6.2), the body shall be customized in reference to model, size, construction, stowage compartments, and other like features.

* 3.5.1 Construction. The vehicle shall be manufactured in accordance with the highest standards of the industry. The body shall be custom built as specified. The body design shall effectively exclude engine fumes and exhaust gases from the interior and shall provide safe and comfortable transportation for passengers and the driver. The vehicle body shall provide a comfortable environment for working personnel for extended periods. The body structure shall be capable of resisting all generally and locally applied loads resulting from normal intended usage including low speed cross country travel, without excessive deflections, cracking, permanent deformation, loosening of joints, or premature fatigue failure. Where aluminum and steel surfaces come into contact, effective means shall be provided to prevent electrolytic action. All equipment and furnishings shall be secured to withstand mobile vehicle operations.

* 3.5.1.1 Structural framing. The body wall, floor and roof shall be of semi-monocoque construction; or the framing shall be 1-1/2 inch aluminum extrusions of nominal 0.125 inch thickness or not less than 1-1/2 inches square tubular aluminum or not less than 1-1/4 inches square tubular high tensile steel. Corners, window frames and door frames shall be reinforced and gusseted to prevent structural distortion. Wood shall not be used for any primary structural framing.

* 3.5.1.2 Structural floor framing. Floor crossmembers shall be adequately sized to support the body and payload without fracture or failure under all combinations of expected loading and shall conform to Federal Motor Carrier Safety Regulation 393.84. Additional crossmembers shall be provided, if required, to support floor loads in critical stress areas.

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* 3.5.1.3 Exterior wall material. The exterior wall skin material shall be either steel, fiberglass or aluminum sheet of sufficient thickness to prevent permanent deformation or buckling. The exterior skin shall be properly attached to the body frame and shall be constructed in a manner to insure rainproof tightness. The exterior skin shall be free of surface roughness or buckling.

* 3.5.1.4 Roof. The roof shall be of one-piece construction with a rise at the center for efficient drainage. The roof shall be properly braced to prevent permanent deformation or buckling, and shall provide a fully weathertight interior. The roof skin shall be either aluminum, fiberglass or a rubber-based material.

* 3.5.1.5 Floor. The floor shall be insulated and shall be sufficiently strong to support the imposed loads under all loading conditions. Galvanized steel, aluminum, fiberglass or other suitable material shall be installed on the underside of the body to fully enclose the floor area. Wood or plywood shall not be exposed to the environment. A sound deadening and vibration damping material shall be applied to the underside of the floor panels exposed to the road environment. The floor covering shall be the manufacturer's standard product.

* 3.5.1.6 Interior panels. The interior of the body, sides, rear and ceiling shall be the manufacturer's standard product. The interior panels shall be dent resistant and properly fastened to the body frame and shall be sufficiently strong to prevent deformation or buckling. Pressed wood shall not be utilized. When specified (see 6.2), the interior panels shall be customized as specified.

* 3.5.2 Insulation. The interior of the body, sides, front, rear and ceiling shall be insulated with a fire retardant material. The insulation shall be installed in such a manner to prevent sag or settling under all operating conditions. The insulation shall not promote the growth of fungus nor cause corrosion of steel or aluminum. The insulation shall provide resistance to the transfer of heat equivalent to the protection afforded by a l-inch thick batt having a thermal conductivity of 0.24 in normal ambient temperatures.

* 3.5.3 Windows and doors. The windows and doors shall be the manufacturer's standard product. The windows shall be provided with fingertip controls permitting easy opening and adjustment to various desired positions. When closed, the windows and doors shall be rainproof and windproof. When specified (see 6.2), the windows and doors shall be customized as to number, size, location, construction and type.

* 3.5.3.1 Windshield. A windshield affording maximum practicable driver vision shall be installed at a slope sufficient to minimize external glare or reflections from inside the body. All exterior glass shall be tinted, heat absorbing safety glass.

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* 3.5.4 Interior furnishings. When specified (see 6.2), the interior furnishings, equipment and accessories required by the procuring activity shall be provided. To facilitate installation, the procuring activity will provide copies of necessary specifications, standards, drawings and publications.

* 3.5.4.1 Interior electrical system. Unless otherwise specified (see 6.2), the interior electrical system, including lighting fixtures, switches, wiring system, interior and exterior receptacles, main power distribution panel, power plant equipment and any other electrical accessories shall be the manufacturer's standard. The electrical system shall be in accordance with the National Fire Protection Association NFPA No. 70 and Federal Motor Carrier Safety Regulations 393.30, 393.31 and 393.33.

* 3.5.4.2 Power plant stowage compartment(s). When specified (see 6.2), an airtight power plant equipment stowage compartment(s) of manufacturer's standard type shall be provided. Complete access to power plant equipment shall be provided. If outside door(s) are provided, the door(s) shall be key-operated. Two sets of keys shall be provided.

3.5.4.3 Heating system. Unless otherwise specified, a heating system of the manufacturer's standard type, shall be furnished. When specified (see 6.2), the heating system shall be as required by the procuring activity.

3.6 Safety. All equipment or exposed positions of the equipment which are subject to extreme temperatures or inclement weather, and all rotating or reciprocating parts which are of such nature or located as to become a hazard to operating personnel, shall be insulated, fully inclosed, shielded or properly guarded.

* 3.6.1 Fire extinguisher(s). When specified (see 6.2), fire extinguisher(s) shall be furnished conforming to A-A-393 of type, class, size and UL rating for class B and C fires. Location and type of bracket(s) as required, shall be furnished.

3.6.2 Highway warning kit. When specified (see 6.2), a highway warning kit, including three emergency reflectors, shall be furnished. The highway warning kit shall conform to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i). The kit shall be stowed in the stowage compartment.

* 3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. All parts shall exhibit no evidence of damage, multilation, or poor workmanship. Damaged components or parts that have been subjected to rebuild or repairs shall not be used as replacement parts or components. All mounting devices such as bolts, nuts and rivets shall be complete and secured in accordance with the highest standards of industry.

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* 3.7.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not degrade the design strength of the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

* 3.7.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts and screws shall be tight.

* 3.7.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of an approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and shall be in full contact with the surface of the member.

* 3.7.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of the parts to be welded shall be free from rust, scale, paint, grease or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loadings.

* 3.7.5 Paneling installation. All interior paneling shall be installed uniformly, free of misalinement, open seams, gaps, and uneven surfaces. Interior paneling shall be properly secured, free of misalined joiners, edgings, waviness, dents, nicks, scratches, sharp edges, and other forms of deformation or surface defects. Exterior paneling shall be free of damaged or missing rivets. Exterior overlapped sheet surfaces shall be sealed with sealing compound with uniform application.

* 3.7.6 Door and window installation. All exterior doors and windows shall be properly located, fitted and sealed to prevent entrance of air, dust and water. All windows and doors with associated hardware shall operate without binding. Window curtains shall be the appropriate color, durable material, free of uneven stitching, crooked seams, runs, tears, or color fade and shall be installed uniformly.

* 3.7.7 Installation of insulation. The insulation shall be installed in a manner excluding excessive voids, open spaces, rips, tears and other types of damage or defects. Wherever practical, the insulation shall be installed prior to the closing of spaces of the body interior. The insulation shall be installed and secured in a manner to retain the material in a permanent position under all operating conditions.

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* 3.7.8 Floor installation and floor covering. The material used to enclose the floor area shall be securely installed and free of gaps, open seams, cuts, or open holes. All wood parts shall be free of cracks or splinters and thoroughly treated to provide full and uniform coverage. The material shall be applied to provide uniform coverage, free of thin or bare areas. The carpeting shall be installed securely and fitted without gaps, visible seams, bulges, frayed edges or color fade.

* 3.7.9 Electrical system installation. The electrical system shall be installed to provide proper routing for electrical wiring with clearances between wiring and applicable parts. All wiring shall be secured and fastened at all points of bending and protected against chafing and damage. Wiring shall be concealed and easily accessible for repairs. All circuits shall be properly grounded. Electrical wiring covering shall be free of cracks, cuts, kinks, frays, or similar defects. All wiring and outlets shall be securely installed, free of loose or faulty connections. Wall outlets and other receptacles shall be conveniently located and installed uniformly. All electrical appliances and fixtures shall be mounted securely and properly positioned to prevent damage during transit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility for ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the

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contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 Treatment of materials. Each vehicle shall be inspected to verify that all ferrous metal used in the construction has been treated and that dissimilar metals have been treated as specified.

4.4 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

* 4.4.1 Vehicle weight. The first production vehicle shall be weighed to determine the curb weight and distribution of the curb weight on the front and rear axles (see 3.2.4 and 3.2.5). The total imposed loads on the front and rear axles shall be computed by the contractor and verified by the Government, using the curb weight, the operator and passenger weight at 175 pounds each, and the payload required to provide the specified GVW. The calculated imposed loads on the front and rear axles shall be compared to the suspension, axle and tire load capacity ratings to determine if these components are of adequate capacity to meet contractual requirements.

* 4.4.2 Road test. To determine conformance to 3.3 through 3.3.1.3, the first production vehicle shall be road tested prior to the water spray test and after completion of the vehicle. The vehicle shall be road tested for a distance of not less than 50 miles. The vehicle shall attain speeds up to at least 55 mph. At least 20 percent of the road test distance shall be over dry, unimproved or gravel roads at speeds up to 20 mph. The vehicle shall be subjected to not less than three sudden stops from speeds of not less than 20 mph on a substantially level, dry, smooth surface free of loose material. The vehicle shall be observed for the specified stopping distance. Upon completion of the road test, the vehicle, components and accessories shall be examined for damage and coolant leakage. All equipment permanently installed or secured for transit shall be examined for damage.

* 4.4.3 In-process water spray test. To determine conformance to 3.5.1.3, 3.5.1.4 and 3.5.3, each vehicle produced under the contract, shall be subjected to a water spray test. The water spray test shall be performed prior to installation of interior paneling and insulation. The vehicle shall be exposed to a 100 percent water spray for not less than 15 minutes. The spray shall be delivered by nozzles operating at not less than 25 pounds per square inch (psi) water pressure. The nozzles shall be sufficient in number and placed approximately 4 feet from the body to afford full coverage of sides, roof, front, rear and undercarriage of the body. The spray shall be applied simultaneously or section by section. All doors and windows shall be closed as in normal travel position. During the water spray test, all windows and paneling shall be

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observed for leakage into the interior of the vehicle. The interior of the underbody storage compartments shall be examined for leakage. Any evidence of leakage (water or moisture) shall be cause for rejection.

* 4.4.4 Truck body treatment and painting. A certification regarding the body cleaning, treating, prime painting and salt spray resistance testing, as required by MIL-STD-1223, shall be made to the Government representatives at the first production vehicle inspection.

* 4.4.5 Radio frequency suppression verification. The vehicle manufacturer shall indicate on the questionnaire (commercial vehicle engineering data) if the vehicle will be suppressed to limit electromagnetic radiation in accordance with SAE J551.

4.4.6 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in a serviceable condition for the duration of the contract.

4.4.7 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.5 Inspection of production vehicles. The contractor's inspection system shall, as a minimum, assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.4.28 have been accomplished.

* 4.5.1 Vehicle body inspection. Vehicle bodies installed on the chassis shall be inspected for the characteristics and defects contained in table II. All noted defects shall be corrected.

TABLE II. Characteristics and defects

Characteristics	Major defects	Method of inspection
Exterior paneling	Unevenness, buckling, warping missing or damaged rivets, overlapped panel surfaces not sealed, excessive or sloppy application or sealing material, gage metal, surface defects (see 3.5.1.3 and 3.7.5)	Visual

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TABLE II. Characteristics and defects. (continued)

Characteristics	Major defects	Method of inspection
Doors and door hardware	Improper fit, not alined, warpage, improper application of door seals, improper material, inoperative door hardware, surface defects (see 3.5.3 and 3.7.6)	Visual and operational
Windows	Sloppy installation of weatherproof seals, windows not easily opened or adjusted to various positions, defective glass (cracks, scratches) (see 3.5.3 and 3.7.6)	Visual and operational
Interior panels	Unevenness, buckling, gaps, open seams, joiners, edgings, and moldings improperly installed, surface defects (see 3.5.1.6 and 3.7.5)	Visual
Floor and floor covering	Gaps, waviness, cracks, bulges, open seams inadequately secured, improper floor covering material, improper application of sound deadening material (see 3.5.1.5 and 3.7.8)	Visual
Stepwells	Hinge cover not flush with interior floor (when applicable), binding or interference of hinged cover or fold-out stair assembly (whichever is applicable) damaged, defective material, improper fit	Visual and functional
Electrical system	Sloppy or loose connections, improper wiring clearances, color code errors, incorrect capacity of circuit breaker box(es), wiring not fastened or clipped, incorrect wire size, faulty materials, faulty solder joints, sloppy installation, inoperative electrical controls, noisy (see 3.5.4.1 and 3.7.9)	Visual and functional

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment from manufacturer's plant to initial receiving activity in accordance with the manufacturer's standard commercial practice.

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6. NOTES

6.1 Intended use. The vehicles covered by this specification are intended for general nontactical use by the Government in transporting personnel or cargo, or for the mounting of special bodies or equipment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
* (b) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Identification of appropriate military service for painting (see 3.1.1.1).
* (d) Identification of appropriate military service for marking (see 3.1.1.2).
(e) Concealed markings, if required (see 3.1.1.2).
(f) Rustproofing, if required (see 3.1.1.3).
(g) Tropical rustproofing, if required (see 3.1.1.3).
(h) Wood treatment, if required (see 3.1.1.5).
(i) Air transportability, if required (see 3.1.1.8).
(j) Lifting and tiedown attachments, if required (see 3.1.1.9).
(k) Blackout curtains, if required (see 3.1.1.10).
(l) Silicone brake fluid, if required (see 3.1.1.12).
(m) Specified wheelbase for body model furnished, if required (see 3.2.7).
* (n) Satisfactory operation on JP-4, JP-5, and JP-8 fuels, if required (see 3.4.1.1).
(o) Gasoline engine, if required (see 3.4.1.2).
* (p) Power plant heaters and fuel warmers, if required (see 3.4.1.8).
* (q) Fuel fired engine preheater, if required (see 3.4.1.9).
* (r) Vernier throttle control, if required (see 3.4.1.10).
* (s) Starting motor circuit breaker, if required (see 3.4.2.1).
* (t) Alternator capacity, if other than as specified (see 3.4.2.3).
(u) Fog lamps, if required (see 3.4.2.8).
(v) Fuel tank(s) of increased capacity, if required (see 3.4.3.2).
* (w) A fuel and water separator, if required (see 3.4.3.3).
(x) Variable rate springs and adjustable shock absorbers, front and rear, if required (see 3.4.8).
(y) Traction control, if required (see 3.4.9.1).
(z) Disc type wheels, if required (see 3.4.10).
* (aa) If other than highway tread tubeless tires are required (see 3.4.10.1).
* (ab) Low profile or bias ply tires, if required (see 3.4.10.1).
(ac) A spare tire carrier, if required (see 3.4.10.3).
(ad) A spare wheel or rim, if required (see 3.4.10.4).
(ae) A spare tire assembly, if required (see 3.4.10.5).
* (af) Intermittent windshield wipers, if required (see 3.4.13).
(ag) Tools, if required (see 3.4.15.1).

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(ah) Standard air conditioning, if required (see 3.4.18).
(ai) An air conditioning system of increased capacity, if required (see 3.4.18) (specify capacity).
(aj) Customized engine compartment, if required (see 3.4.19) (specify details).
* (ak) Automatic engine shutdown system, if required (see 3.4.21).
* (al) Remote control curbside rear view mirror, if required (see 3.4.22).
* (am) Heated rear view mirrors. if required (see 3.4.22).
(an) Additional accessories, if required (see 3.4.23).
(ao) Insect screen, if required (see 3.4.25).
(ap) Leveling jacks, if required (see 3.4.26).
(aq) An engine hour meter, if required (see 3.4.27).
* (ar) Cooling system protection down to -65°F, if required (see 3.4.28.1).
(as) Customized body, if required (see 3.5) (specify details).
(at) Customized interior panels, if required (see 3.5.1.6) (specify details).
(au) Customized windows and doors, if required (see 3.5.3) (specify details).
(av) Customized interior furnishings, if required (see 3.5.4) (specify details).
(aw) Customized interior electrical system, if required (see 3.5.4.1) (specify details).
(ax) Power plant stowage compartment(s), if required (see 3.5.4.2) (specify details).
(ay) Customized heating system, if required (see 3.5.43) (specify details).
(az) Fire extinguisher(s), if required (see 3.6.1).
(ba) Highway warning kit, if required (see 3.6.2).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type and class. Unless other conditions are cited in this contract, computations should be made for normal atmospheric pressure, normal ambient ait temperature, and still, dry, air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE J688, Truck Ability Prediction Procedure Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00

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Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW and the engine revolutions per minute (rpm) (to nearest 100 revolutions) which is required for a geared speed of 50 mph. For engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

6.4 Subject term (keyword) listing.

Customized Travel Unit
Forward Control Truck
Nontactical Truck
Truck, Van
4x2.

6.5 Changes from previous issue. The margins of this specification are marked with asterisks (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Army – AT
Air Force - 99

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